

NARENDRA VANAPALLI

AI ML & Full Stack Engineer

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PROFESSIONAL SUMMARY

Full Stack and AI/ML Engineer with **2 years of experience** building scalable, user-focused web applications and AI-powered enterprise solutions. Hands-on experience across the full development lifecycle using **ReactJS, Python/FastAPI, Java/Spring Boot**, and **Spring WebFlux**, with expertise in designing **RESTful APIs** and working with **MongoDB, PostgreSQL, Redis**, and caching solutions. Experienced in cloud-native application development, containerization with **Docker**, and collaborative delivery using **Git/GitHub**. A **Databricks Certified Generative AI Engineer Associate** and **AWS Certified AI Practitioner**, with experience developing **Generative AI, RAG**, and **agentic AI** solutions, including responsible AI practices for secure, governed, and reliable enterprise workflows.

TECHNICAL SKILLS

Programming & Backend: Python, Java, JavaScript, FastAPI, Spring Boot, Spring WebFlux, Flask, RESTful APIs, Microservices.

Frontend: ReactJS, Redux, Bootstrap, HTML5, CSS3, Responsive Web Design.

Databases & Caching: MongoDB, PostgreSQL, Microsoft SQL Server, Redis, Caffeine Cache, SQLAlchemy, pgvector.

Cloud & DevOps: AWS, Amazon Bedrock, Docker, CI/CD, Git, GitHub, Maven, Postman, Swagger.

Testing: PyTest, JUnit, Unit Testing, Integration Testing, Regression Testing.

Generative AI: Generative AI, LLMs, RAG, Agentic AI, LangChain, LangGraph, LangSmith, OpenAI API, AI Evals, Responsible AI, AI Guardrails.

AI Development & Practices: OpenAI Codex, Claude Code, GitHub Copilot, Windsurf, GitHub Spec Kit, Spec-Driven Development.

Development Tools: VS Code, Eclipse, STS, IntelliJ IDEA.

PROFESSIONAL EXPERIENCE

Endava Solutions India Private Limited

Aug 2024 – Present

Machine Learning Engineer (Promoted: Associate → Developer → ML Engineer)

International Technology Park Ltd., Bangalore

- Delivered **5 full-stack and AI-powered applications** across healthcare, automation, and AI domains using ReactJS, Python/FastAPI, Java/Spring WebFlux, MongoDB, PostgreSQL, and Redis.
- Engineered an **Agentic AI Companion and Conversational Business Intelligence Platform** using React, FastAPI, and AWS Bedrock AgentCore, integrating RAG, multi-agent workflows, external REST APIs, natural-language analytics, and interactive dashboards.
- Applied Redis-backed short- and long-term memory, semantic caching, response caching, and concurrent-request coalescing, lowering repeated analytical-request latency by approximately **85%**.
- Created the **Automated Regression Validator**, a full-stack data-validation application using ReactJS, Python/FastAPI, MongoDB, and Pandas, supporting configurable validation rules, cross-run issue tracking, data-quality analysis, and dashboard reporting.
- Created the **Automated Regression Validator**, a full-stack data-validation application using **ReactJS, Python/FastAPI, MongoDB**, and **Pandas**; automated **30+ validation checks** across **1M+ records**, lowering manual validation time by approximately **60%**.
- Launched **Endava Talent Sphere**, an internal employee-management and workforce-insights application using ReactJS, Java Spring WebFlux, PostgreSQL, and Caffeine caching; crafted responsive interfaces, non-blocking APIs, and employee-data workflows.
- Accelerated API, database, and application performance through query tuning, indexing, asynchronous processing, Redis caching, and connection management, decreasing response latency by approximately **87%**, from **3 seconds to 400 milliseconds**.
- Partnered with engineers, product stakeholders, and designers across Agile ceremonies, code reviews, debugging, and end-to-end delivery of full-stack and AI solutions.

GalaxE Solutions, Inc.

Oct 2023 – Feb 2024

Software Engineering Intern

Bangalore, India

- Completed hands-on software-engineering training in **Java, Spring Boot, ReactJS, PostgreSQL, Git**, RESTful APIs, authentication, and full-stack development practices.
- Developed a full-stack **MuscleBlaze e-commerce store** with responsive ReactJS user interfaces and Spring Boot backend services for product catalog management, search, cart operations, user accounts, orders, and checkout workflows.
- Designed RESTful APIs, normalized PostgreSQL schemas, JWT-based authentication, role-based access control, and caching mechanisms to support secure and responsive frontend-to-backend application workflows.
- Applied Git-based collaboration, debugging, testing, and code-review practices to deliver an end-to-end e-commerce application within the training program timeline.
- Recognized as the **top performer among 120+ trainees** for technical proficiency, project delivery, and full-stack development capability.

FEATURED PROJECTS

Agentic AI Companion & Conversational Business Intelligence Platform

Enterprise Healthcare Client | Apr 2026 – Jun 2026

- Engineered RESTful APIs and multi-agent workflows for RAG-based retrieval, natural-language-to-SQL analytics, external API execution, and dashboard generation.
- Configured Redis-based short-term and long-term memory, semantic caching, response caching, and concurrent-request coalescing.
- Impact:** Cut repeated analytical-request latency by approximately **80%**, improving responsiveness for follow-up analytics queries.

Endava Talent Sphere – Employee Management Platform
Internal Product, Endava | Jan 2025 – Jun 2026

- Built an employee-management application using **ReactJS, Java Spring WebFlux, PostgreSQL, REST APIs**, and **Caffeine** caching.
- Created responsive interfaces and non-blocking backend services for employee information, organizational hierarchies, and workforce insights.
- Released the application internally through CI/CD pipelines, coordinating frontend, backend, and database integration.
- **Impact:** Centralized employee-data access and workforce-management workflows for **5000+ internal users**.

Product Match AI – Competitor Intelligence Platform
Global Medical Device Client | Aug 2025 – Oct 2025

- Delivered a full-stack medical-device competitor-intelligence platform using **ReactJS, Python/FastAPI, PostgreSQL, pgvector**, and RESTful APIs.
- Engineered a multi-stage AI matching pipeline combining vector search, metadata filtering, cross-encoder reranking, and LLM-generated explanations.
- Designed product-upload, search, match-review, progress-tracking, and structured-export workflows.
- **Impact:** Decreased analyst effort and time spent identifying relevant competitor products by approximately **50%**.

Automated Regression Validator (ARV)
Internal Product, Endava | Jun 2025 – Aug 2025

- Developed a full-stack data-validation application using **ReactJS, Python/FastAPI, MongoDB, Pandas**, and RESTful APIs.
- Created configurable validation workflows for missing values, invalid formats, duplicate records, business-rule violations, and migration-readiness issues.
- Added regression comparison capabilities to identify new, resolved, and recurring data-quality issues across validation runs.
- **Impact:** Automated **30+ data-quality checks** across **1M+ records**, reducing regression-validation time by approximately **60%**.

AWARDS & ACHIEVEMENTS

- Topped the **Codex Hackathon** for developing **Dava Compass**, an AI-assisted workforce-planning and staffing decision platform.
- Secured **2nd place in a regional AI innovation competition** for developing **AI Evaluator**, an LLM-based code-assessment platform **[Post]**.
- Earned the **Databricks Certified Generative AI Engineer Associate** and **AWS Certified AI Practitioner** certifications.
- Completed **Reactive Programming in Modern Java using Project Reactor** training through Udemy, with hands-on learning in **Spring WebFlux** and reactive application development.
- Promoted from **Junior Developer** to **Machine Learning Engineer** within **2 years**, reflecting consistent technical growth and contribution across full-stack and AI projects.
- Completed certified training in **Generative AI, LangChain/LangGraph, Agentic AI, Data Science**, and **Machine Learning**.
- Recognized as the **top performer among 120+ trainees** during the GalaxE Solutions Software Engineering training program for full-stack project delivery and technical proficiency.

EDUCATION

University College of Engineering Kakinada, JNTUK
Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

2020 – 2024
Kakinada, India